



Data Analysis Report

SALES ANALYSIS

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COMPANY INTRODUCTION

NovaTech Electronics is a growing consumer electronics retailer, operating across multiple cities in the United States. The company sells products such as laptops, smartphones, monitors, headphones, batteries, and electronic accessories through online sales channels.

With increasing sales volume and seasonal demand fluctuations, the company aims to use data-driven insights to improve sales performance, inventory planning, marketing strategies, and overall business growth.

PROBLEM

- ❖ Sales performance declines significantly during **mid-year months**, creating inconsistent revenue growth throughout the year.
- ❖ Revenue is heavily dependent on a small number of **premium products**, increasing business risk and revenue concentration.
- ❖ **Inventory shortages** during peak shopping periods may lead to **missed sales opportunities** and delayed order fulfillment.
- ❖ Customers frequently **purchase accessories together with electronics**, but the company is not fully utilizing bundling and cross-selling opportunities.
- ❖ Marketing campaigns are not optimized around **peak shopping hours, seasonal demand, and high-performing cities**.
- ❖ Some cities generate significantly lower sales performance & require **targeted marketing strategies** to improve growth.

GOAL

The objective of this analysis is to identify revenue trends, customer purchasing behavior, product performance, city-level sales patterns, and peak demand periods in order to improve inventory planning, marketing strategies, cross-selling opportunities, and overall business performance.

KEY BUSINESS QUESTIONS

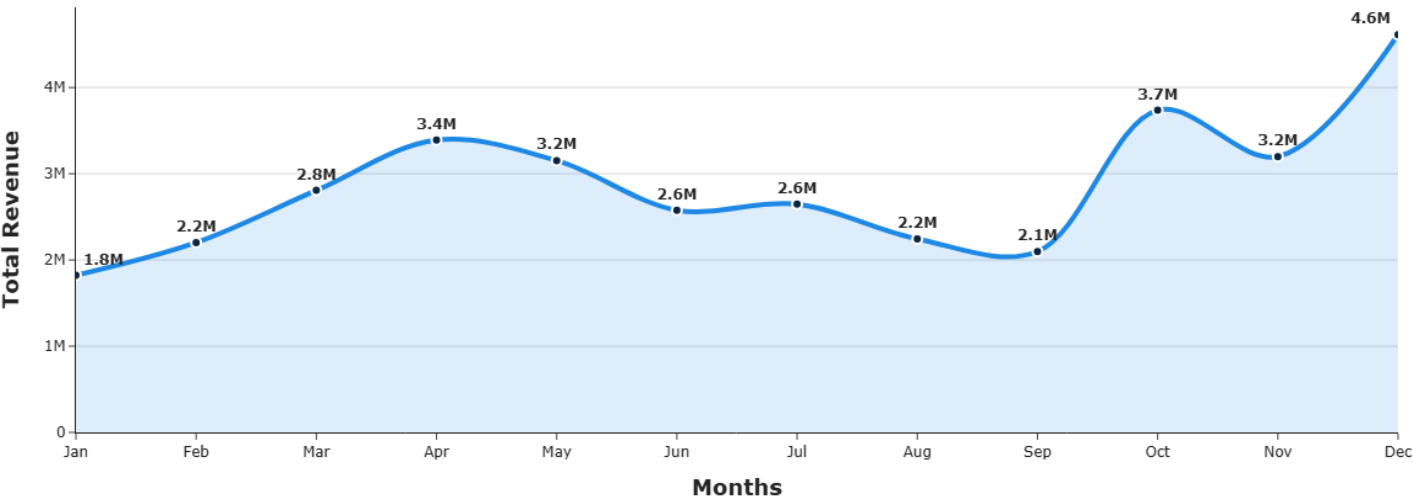
- 1. Revenue Performance
- 2. Product Performance
- 3. Revenue Concentration Risk
- 4. Time-Based Sales Behavior
- 5. City Performance
- 6. Market Basket Analysis
- 7. Time-Based Product Demand
- 8. Business Recommendations(Suggestions)

KEY FINDINGS

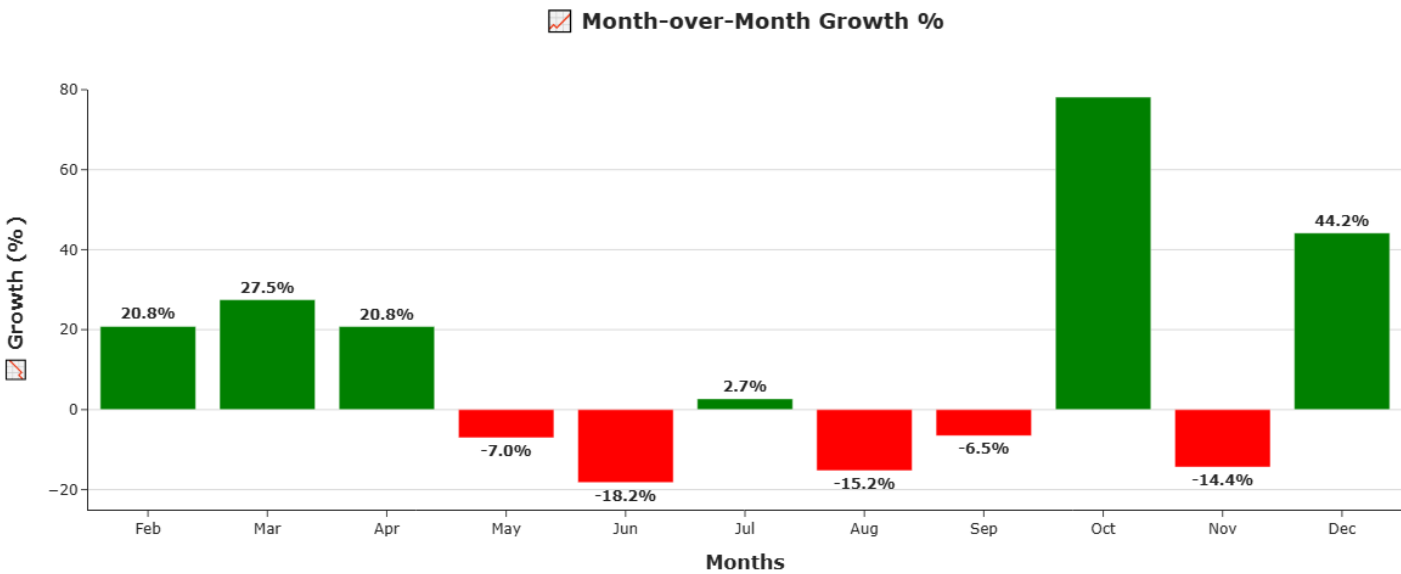
OVERALL PERFORMANCE

- Total Revenue: **\$34,492,036**

Sales Revenue Trend by Month



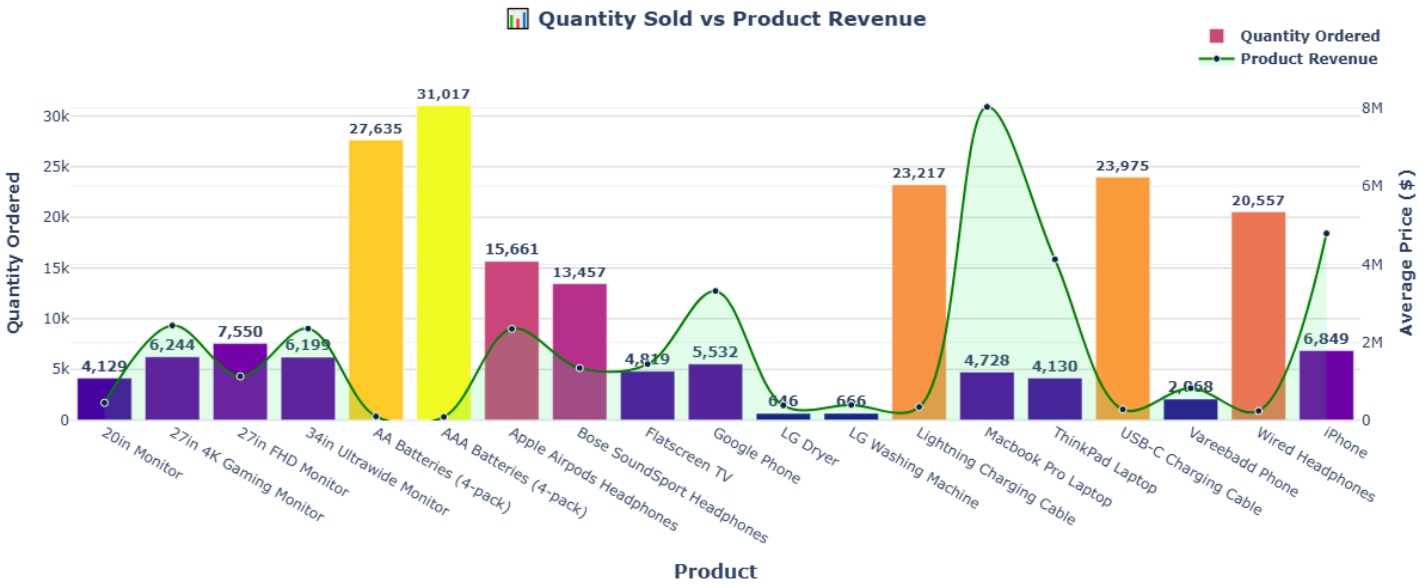
Month	Total Revenue	Growth (%)
Dec	\$4,613,443	44.19%
Oct	\$3,736,727	78.15%
Apr	\$3,390,670	20.79%
Nov	\$3,199,603	-14.37%
May	\$3,152,607	-7.02%
Mar	\$2,807,100	27.48%
Jul	\$2,647,776	2.71%
Jun	\$2,577,802	-18.23%
Aug	\$2,244,468	15.23%
Feb	\$2,202,022	20.84%
Sep	\$2,097,560	-6.55%
Jan	\$1,822,257	UnKnown



- Best Month: **Oct (78.15%)**
- Worst Month: **Jun (-18.23%)**

PRODUCT PERFORMANCE

Product	Total Revenue	Quantity Ordered
Macbook Pro Laptop	\$8,037,600	4,628
iPhone	\$4,794,300	6,729
ThinkPad Laptop	\$4,129,959	4,038
Google Phone	\$3,319,200	5,423
27in 4K Gaming Monitor	\$2,435,098	6,118
34in Ultrawide Monitor	\$2,355,558	6,072
Apple Airpods Headphones	\$2,349,150	15,016
Flatscreen TV	\$1,445,700	4,719
Bose SoundSport Headphones	\$1,345,565	12,894
27in FHD Monitor	\$1,132,424	7,393
Vareebadd Phone	\$827,200	2,030
20in Monitor	\$454,149	4,049
LG Washing Machine	\$399,600	653
LG Dryer	\$387,600	633
Lightning Charging Cable	\$347,094	21,715
USB-C Charging Cable	\$286,501	22,109
Wired Headphones	\$246,478	19,120
AA Batteries (4-pack)	\$106,118	27,136
AAA Batteries (4-pack)	\$92,741	30,372

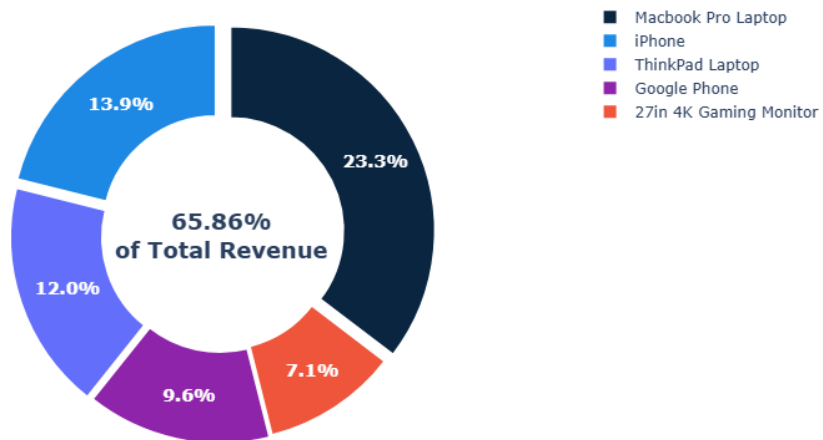


REVENUE CONCENTRATION RISK

Product	Total Revenue	Revenue Percentage (%)
Macbook Pro Laptop	\$ 8,037,600	23.3 %
iPhone	\$ 4,794,300	13.9 %
ThinkPad Laptop	\$ 4,129,959	12.0 %
Google Phone	\$ 3,319,200	9.6 %
27in 4K Gaming Monitor	\$ 2,435,098	7.1 %

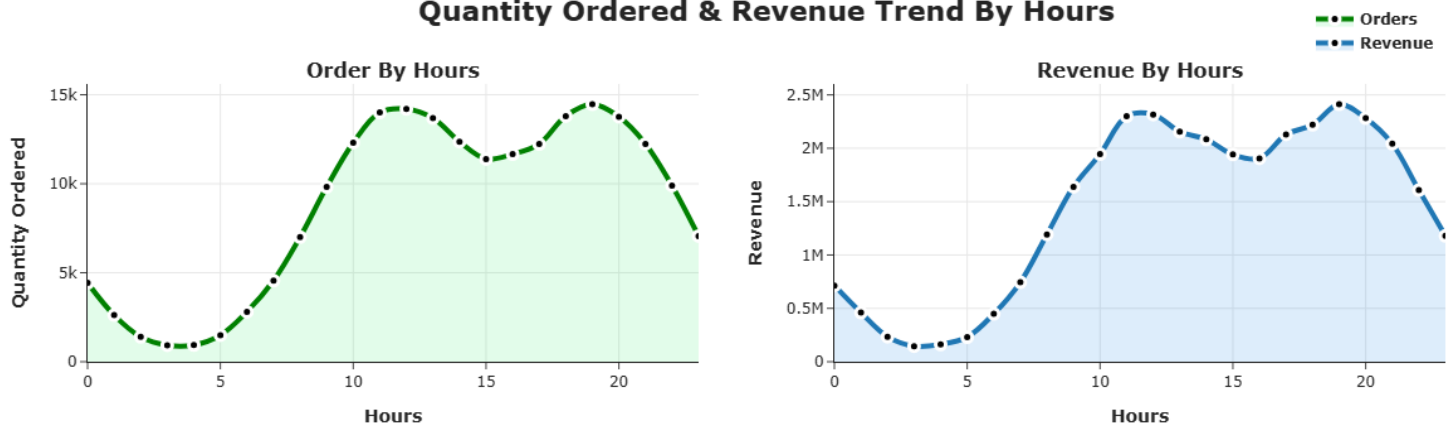
- Top 5 products generate: **65.86 %** of Total Revenue.

Revenue Concentration – Top 5 Products



TIME-BASED SALES BEHAVIOR

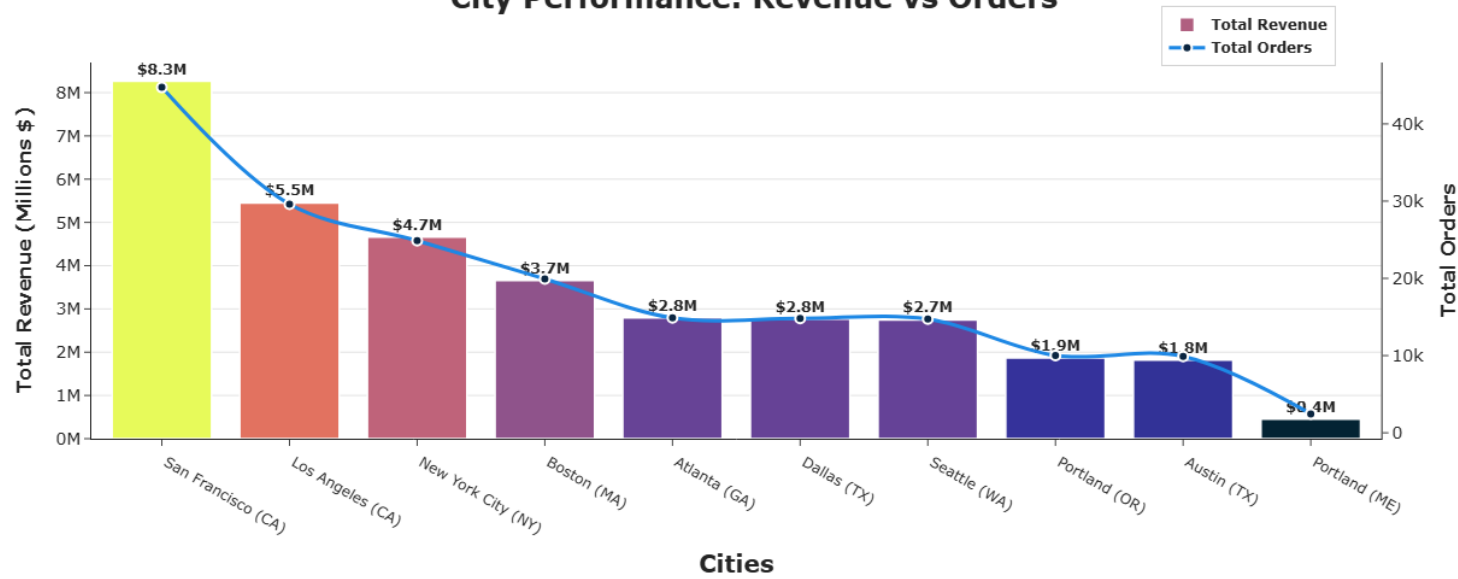
Quantity Ordered & Revenue Trend By Hours



CITY PERFORMANCE

City	Total Revenue	Quantity Ordered
San Francisco (CA)	\$ 8,262,203	44,732
Los Angeles (CA)	\$ 5,452,570	29,605
New York City (NY)	\$ 4,664,317	24,876
Boston (MA)	\$ 3,661,642	19,934
Atlanta (GA)	\$ 2,795,498	14,881
Dallas (TX)	\$ 2,767,975	14,820
Seattle (WA)	\$ 2,747,755	14,732
Portland (OR)	\$ 1,870,732	10,010
Austin (TX)	\$ 1,819,581	9,905
Portland (ME)	\$ 449,758	2,455

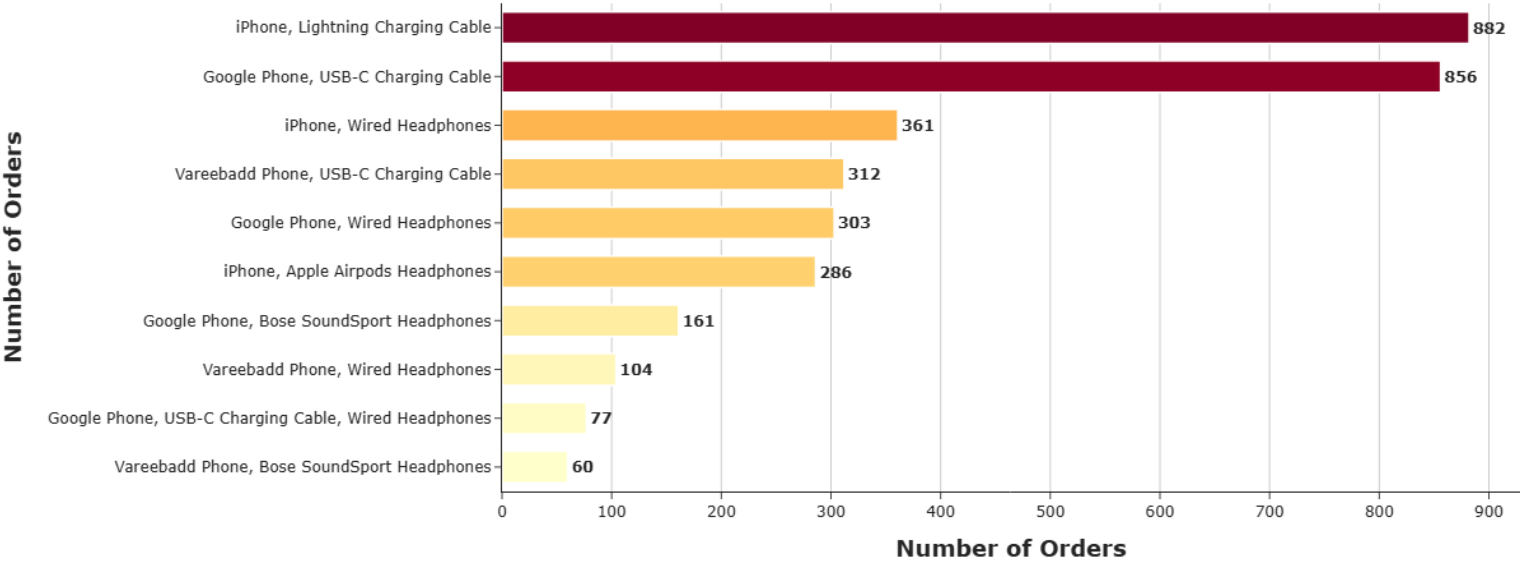
City Performance: Revenue vs Orders



MARKET BASKET ANALYSIS

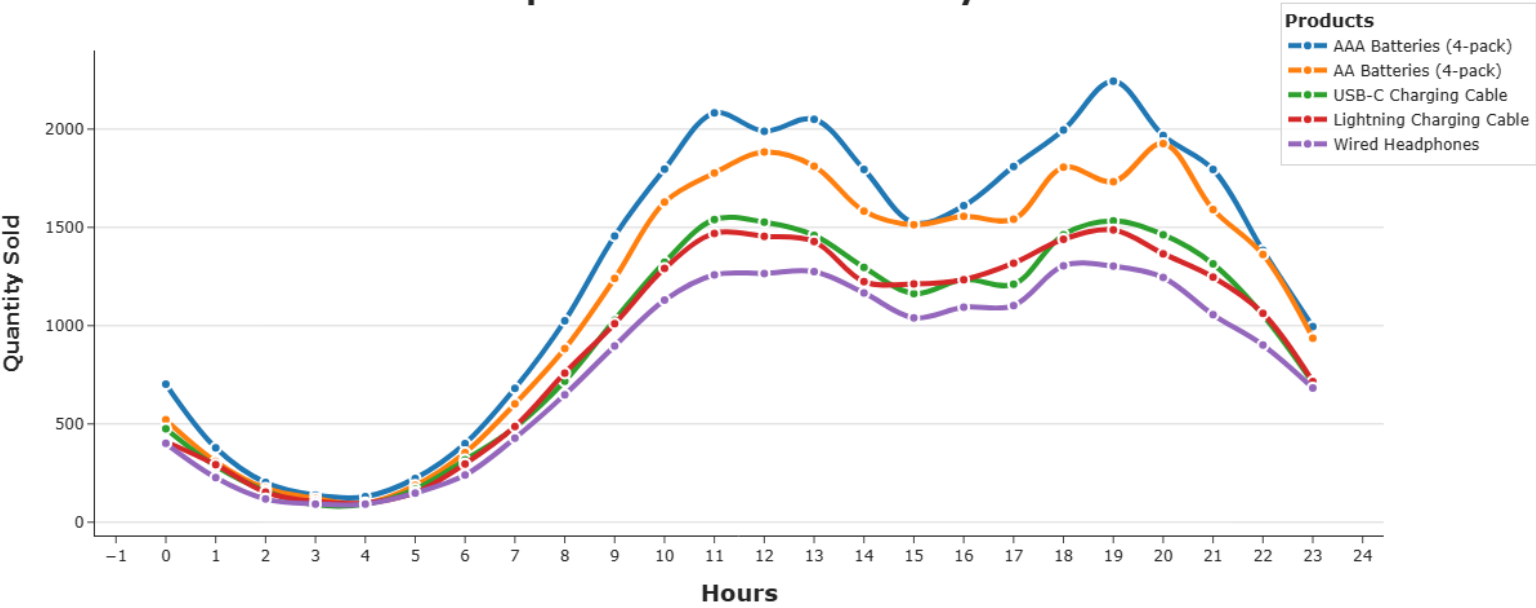
Products Sold Together	Count
iPhone, Lightning Charging Cable	882
Google Phone, USB-C Charging Cable	856
iPhone, Wired Headphones	361
Vareebadd Phone, USB-C Charging Cable	312
Google Phone, Wired Headphones	303
iPhone, Apple Airpods Headphones	286
Google Phone, Bose SoundSport Headphones	161
Vareebadd Phone, Wired Headphones	104
Google Phone, USB-C Charging Cable, Wired Head...	77
Vareebadd Phone, Bose SoundSport Headphones	60

Top Products Frequently Purchased Together

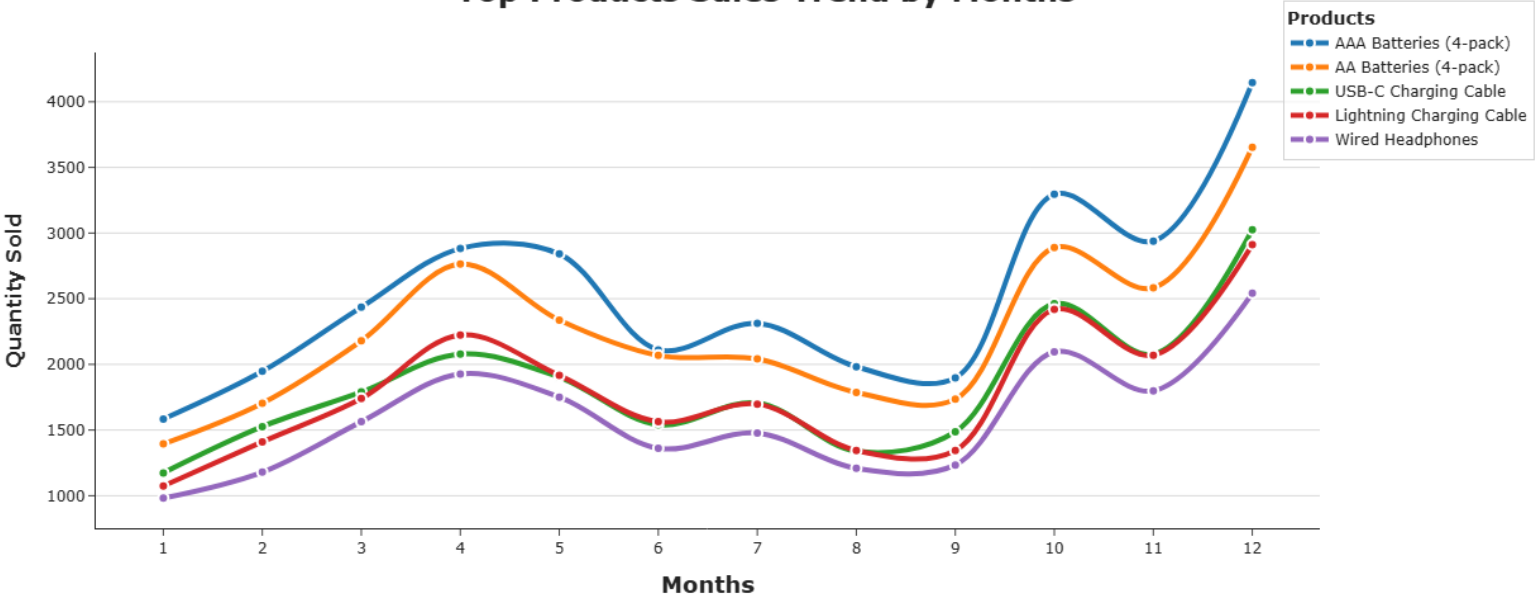


TIME-BASED PRODUCT DEMAND

Top Products Sales Trend by Hour



Top Products Sales Trend by Months



BUSINESS RECOMMENDATIONS(SUGGESTIONS)

REDUCE REVENUE CONCENTRATION RISK

The top 5 products contribute approximately 65.86% of total revenue, which creates a high business dependency risk. If demand for one major product drops, overall company revenue could decline significantly.

Recommendation:

- Expand marketing campaigns for low-performing products such as cables, headphones.
- Create product bundles with high-demand devices (e.g., MacBook + accessories).
- Introduce cross-selling strategies during checkout.

INCREASE SALES DURING LOW-REVENUE MONTHS

Sales performance declines noticeably during May, June, August, and September, while October and December generate peak growth.

Recommendation:

- Launch seasonal campaigns and limited-time discounts during slow months.
- Run mid-year promotions, back-to-school campaigns, and bundle offers.
- Increase digital advertising spend before weak-performing months.

OPTIMIZE SALES AROUND PEAK HOURS

The highest order volume and revenue occur between 11 AM – 8 PM, especially around 7 PM.

Recommendation:

- Schedule paid advertisements and email campaigns during peak conversion hours.
 - Ensure inventory and customer support availability during high-demand periods.
 - Use flash sales during peak traffic windows.
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IMPROVE PERFORMANCE IN LOW-REVENUE CITIES

Cities like Portland (ME) and Austin (TX) generate significantly lower revenue compared to San Francisco and Los Angeles.

Recommendation:

- Conduct localized marketing campaigns in underperforming cities.
- Offer city-specific promotions and free shipping incentives.
- Analyze regional customer preferences and product demand.

LEVERAGE MARKET BASKET OPPORTUNITIES

Customers frequently purchase products together, such as:

1. iPhone + Lightning Charging Cable
2. Google Phone + USB-C Charging Cable
3. iPhone + Wired Headphones

Recommendation:

- Introduce “Frequently Bought Together” recommendations on the website.
- Offer bundle discounts for commonly paired products.
- Use AI-based recommendation systems for personalized upselling.

IMPROVE INVENTORY PLANNING

Products such as batteries and charging cables sell in very high quantities but generate relatively lower revenue individually.

Recommendation:

- Maintain higher inventory levels for fast-moving accessories.
 - Use demand forecasting models for stock optimization.
 - Prevent stockouts during peak sales hours and months.
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